Practical No. 10

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code :

```
  GNU nano 8.7.1                                                                          sstf.c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total_movement = 0;
    int visited[8] = {0};

    printf("Seek Sequence: ");

    for(int i = 0; i < n; i++)
    {
        int min = 1000;
        int index = -1;

        for(int j = 0; j < n; j++)
        {
            if(!visited[j])
            {
                int distance = abs(head - requests[j]);

                if(distance < min)
                {
                    min = distance;
                    index = j;
                }
            }
        }

        total_movement += min;
        head = requests[index];
        visited[index] = 1;

        printf("%d ", head);
    }

    printf("\nTotal Head Movement: %d\n", total_movement);

    return 0;
}
```

Output :

```
~
PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ nano sstf.c

PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ gcc sstf.c -o sstf

PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ ./sstf
Seek Sequence: 41 65 67 98 122 124 183 14
Total Head Movement: 323

PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ |
```